LESSON 105 (1.5 GROUND and 1.3 DUAL)

Lesson Objective

During this lesson, the student will conduct a preflight inspection in preparation for flight and apply the techniques practiced in the training device in the aircraft. The student will increase his/her understanding of the effects of the primary, secondary, and ancillary aircraft controls and demonstrate that in flight.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
  - Quiz the student on primary and secondary flight controls.
  - Review the passenger briefing, engine fire during start briefing, and the takeoff briefing.
  - Quiz the student on the readings as needed.
    - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
      - The four fundamentals.
      - Effects and use of flight controls.
      - Attitude flying.
      - Straight and level flight.
- Student questions.

References

- Advisory Circulars. (Search Tool)
  - Pilots' Role in Collision Avoidance. (AC 90-48)
- Airplane Flying Handbook. (FAA-H-8083-3C)
  - Chapter 03 - Basic flight maneuvers.
- Pilot's Handbook of Aeronautical Knowledge. (FAA-H-8083-25B)
  - Chapter 02 - Aeronautical Decision-Making.
  - Chapter 06 - Flight Controls.
- Warrior III Pilot's Information Manual.
  - Section 06 - Weight and Balance.
  - Section 07 - Description and operation of the airplane and its systems.

Lesson Content (Ground)

Review Fundamentals from Lesson 104

- Effects of controls, power, flaps, etc., from AATD to be orally quizzed.

Review Weather Briefing

- Quiz the student on:
  - Official vs non-official weather sources.
  - Different ways to obtain a weather briefing.

    - Types of weather briefings.
- Have the student brief the current weather pertaining to the flight.

Review Performance & Weight and Balance (TOLD card)

- Review and inspect the student's TOLD card.
- Ask the student questions on:
    - Weight and balance.
    - Pressure altitude and density altitude.
    - Takeoff and landing distance.
    - V-speeds.

Primary and Secondary Aircraft Controls

- Quiz the student on primary and secondary aircraft controls.

Effects of Flight Controls

- Primary:
    - Elevator → Pitch.
    - Rudder → Yaw.
    - Ailerons → Roll.
- Secondary:
    - Elevator → Airspeed.
    - Rudder → Roll.
    - Ailerons → Yaw.
- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
    - Faster airflow will result in a more responsive flight control.
    - Slower airflow will result in a less responsive flight control.

Operation of Systems - Landing Gear, Fuel, Oil, and Hydraulic

- Warrior III Pilot's Information Manual section 7.
- Landing Gear:
    - Tricycle type, fixed landing gear.
        - Single disc double puck hydraulic brakes located on each main gear.
    - Struts:
        - Oleo type:
            - Combination of air (Nitrogen) and oil.
            - Main struts should be extended 4.50” and the nose 3.25” - PIM Section 8.
        - Nose strut has a shimmy dampener to avoid any unwanted vibrations on takeoff or landing.
    - Tires:
        - Mains - 24 psi for MGTOW.
        - Nose - 30 psi for MGTOW.

- Brakes:
  - Warrior has 5 hydraulic cylinders, one for each toe brake and one for the parking brake.
  - Hydraulic fluid is transferred through the hydraulic lines by depressing the brake pedals.
  - This moves the brake pads and creates friction to slow the aircraft down while on the ground.
  - Hydraulic fluid is red (MIL-H-5606) - PIM Section 8.
- Nose Wheel Steering:
  - Steering on the ground is accomplished through the use of the rudder pedals.
  - Rudder pedals are directly linked to the nose wheel.
  - Brakes may also be used while on the ground to aid in turning.

- Fuel System:
  - General:
    - 50 gallons total.
    - 48 usable - 24 per side.
    - 34 gallons usable if filled to the bottom of the tabs - 17 per side.
  - Components:
    - Fuel selector:
      - Left side of cockpit near floor.
      - 3 settings: off, right, left.
    - Gascolator:
      - Lowest point in the fuel system.
      - This is where we can find deposits in the fuel system.
    - Electric fuel pump:
      - Mounted to the firewall.
      - Serves as a backup to the engine driven fuel pump.
      - Used during takeoff, maneuvers, and landings as a “boost” to the engine driven pump.
    - Engine driven fuel pump:
      - Mounted to the outside of the engine.
    - Fuel quantity gauges.
    - Fuel flow/fuel pressure gauge.
    - Fuel nozzles:
      - Fuel flows from the regulator continuously into the nozzle through a calibrated orifice.
      - Fuel collects in the chamber at the center of the nozzle when the intake valve is closed, and when open low pressure pulls the fuel and air into the intake port just before the cylinder.
- Oil System:
  - 2 types:

        - Mineral oil:
            - Used for the first 50 hours of new or overhauled engines.
            - Allows for a faster and more complete seal~~t~~ing of the piston rings and other internal parts.
        - Ashless dispersant:
            - Used after 50 hours.
            - Prevents development of carbon and ash deposits in the engine.
            - Keeps contaminants in the oil, allowing them to be trapped in the oil filter.
    - Components:
        - Oil stored in a wet-type oil sump beneath the engine crankshaft.
        - Up to 8 quarts of SAE 50 W oil → FIT 6 quarts. PIPER 2 quarts.
            - SAE: Society of Automotive Engineers.
            - 50 W: viscosity, or weight, of oil.
        - Oil pump located inside the engine.
        - Oil cooler is mounted outside of the engine.
        - Oil filter is mounted outside of the engine.
    - Movement:
        - Sump – suction screen – oil pump – bypass valve – oil cooler – oil filter – engine and propeller – sump.
    - 4 purposes:
        - Clean.
        - Cool.
        - Lubricate.
        - Seal.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic when weather permits.
- FAR regulation 91.111 - operating near other aircraft.
    - Cannot operate near another aircraft to create a collision hazard.
    - Cannot fly in formation unless there was a prior agreement between the PIC's of participating aircraft.
    - No passengers for hire in formation flight.
- Explain how to scan for traffic in daytime conditions.
    - 10 degrees per second, looking both above and below aircraft level.
- Clearing procedures:
    - During taxi.
    - Before climbs.
    - During descents.
    - Clearing turns.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Preflight Inspection

- For the first flight, it is recommended that the instructor supervises the student's preflight.

Go/No-Go Decision

- The student must make a go/no-go decision depending on:
    - Personal minimums.
    - FIT minimums.
    - Aircraft.
    - Airspace.
- Review the students planning with a focus on:
    - Weather, weight and balance, performance, flight planning, and fuel calculations.
    - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
    - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
    - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

<u>**Lesson Content (Dual)**</u>

Checklist Procedures

- <u>Re-emphasize the importance of using checklists.</u>
- Have the student run through all of the checklists.
- Assist if needed.
- Have the student go through the "passenger briefing" and the "engine fire during start brief".

Preflight Inspection

- For the first flight, it is recommended that the instructor supervises the student's preflight.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.
- Demonstrate how to scan for traffic in daytime conditions.
    - 10 degrees per second, looking both above and below aircraft level.
- IF YOU SEE SOMETHING, SAY SOMETHING!

Primary Effects of Flying Controls

- <u>Emphasize to the student that all control movements must be smooth!</u>
- Elevator → Pitch.
- Rudder → Yaw.
- Ailerons → Roll.

Secondary Effects of Flying Controls

- Elevator → Airspeed.
- Rudder → Roll.
- Ailerons → Yaw.

- Introduce the concept of a spiral dive and how to properly recover.

Effects of Airspeed

- The speed that air is moving over a flight control surface will greatly impact the effectiveness of said surface.
    - Faster airflow will result in a more responsive flight control.
    - Slower airflow will result in a less responsive flight control.

Effect of Slipstream

- Fast.
    - Elevator → More Pitch.
    - Rudder → More Yaw.
    - Ailerons → More Roll.
- Slow.
    - Elevator → Less Pitch.
    - Rudder → Less Yaw.
    - Ailerons → Less Roll.

Effect of Trim

- <u>Emphasize the importance of trim!</u>
- Natural sense.
    - Roll wheel forward for forward trim.
    - Roll wheel aft for aft trim.
- Coarse versus fine trim.
- Pitch to select attitude, trim to relieve pressure.
- Any adjustment to pitch or power should be followed with an adjustment to trim.

Engine Handling

- Highlight the red line on the RPM gauge.
- Ensure that the student understands the negative effects of "redlining" the engine.
- When stationary on the ground, RPM should be set to 1000 RPM.
- Do not lean the mixture below 3,000 ft.
    - For leaning guidance, refer to the PIM and the leaning guidance on the FIT hub.

Effect of Power and Airspeed:

- For an increase in power.
    - The pitch will increase and there will be a yaw to the left.
    - To correct you need forward pressure on the yoke and right rudder.
- For a decrease in power.
    - The pitch will decrease and there will be a yaw to the right.
    - To correct you need back pressure on the yoke and left rudder.
- For an increase in airspeed.
    - The pitch will increase and there will be a yaw to the right.
    - To correct you need forward pressure on the yoke and left rudder.

- For a decrease in airspeed.
  - The pitch will decrease and there will be a yaw to the left.
  - To correct you need back pressure on the yoke and right rudder.

Effect of Flaps

Flap Setting / Effects	Lift	Nose	Elevator (initially)	Drag	Airspeed	Elevator (due to drag)
0 - 10 Degrees	↑↑↑	↑↑	Forward	↑	↓	Backward
10 - 25 Degrees	↑↑	↑↑	Forward	↑↑	↓↓	Backward
25 - 40 Degrees	↑	↑	Forward	↑↑↑	↓↓↓	Backward
40 - 25 Degrees	↓	↓	Backward	↓↓↓	↑↑↑	Forward
25 - 10 Degrees	↓↓	↓↓	Backward	↓↓	↑↑	Forward
10 - 0 Degrees	↓↓↓	↓↓	Backward	↓	↑	Forward

Operation of Mixture Control

- The mixture lever has a red knob and is located to the right of the throttle lever.
- Moving the mixture lever forward results in a rich fuel/air mixture, while moving the mixture lever backward results in a lean fuel/air mixture.

Operation of Carburetor Heat

- The carburetor heat lever is located to the right of the mixture lever.
- To operate, pull the lever downward to the on position or upward to the off position.
- Review when carburetor heat is required (show in checklists).

Operation of Environmental Controls

- Show where air vents are located in the aircraft, and how to operate.
- Demonstrate how to operate the cabin heat and defroster.

Other Controls as Applicable

- Adjusting audio panel power selection, volume, squelch, Garmin430 basic usage (com selection, frequency changes, power selection).

Postflight Procedures

- For the first flight, it is recommended that the instructor supervises the postflight procedures.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.

- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 106.
- Assign homework:
    - Review the landing gear, hydraulic, fuel, and oil systems.
    - Review taxiing procedures in the Warrior Maneuvers' Manual.
    - Review chapter 03 of the Airplane Flying Handbook - Basic Flight Maneuvers:
        - The four fundamentals.
        - Effects and use of flight controls.
        - Attitude flying.
        - Straight and level flight.

Completion Standards

This lesson is complete when the student can:

1. Have a weather briefing and completed TOLD card prepared.
2. Demonstrate an understanding of checklist procedures.
3. Demonstrate the effects of primary, secondary and ancillary controls in the airplane with little instructor guidance.